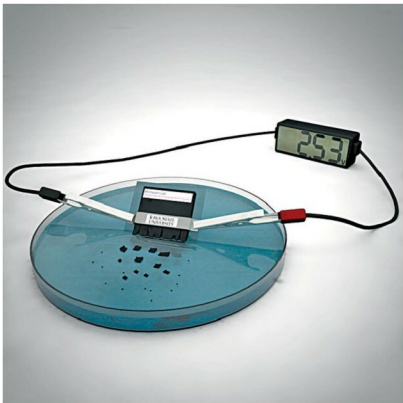


Battery that self-destructs in 30 minutes

Iowa State University, USA, scientists have developed a self-destructing battery that can power a simple electronic device such as a four-function calculator for up to 15 minutes and then dissolve in water. This could pave the way for so-called transient power sources for scientific instruments or tools of espionage.

This new battery represents a marked improvement in voltage and disintegration time over its predecessors, according to the researchers. The battery's polymer casing is made from a molecule that can form long repeating chains, swell and physically break itself and other components into small pieces when exposed to water. Devices powered by this type of battery could serve their function or transmit data and then be washed away in the rain.

The battery can generate about 2.7 volts, which is similar to the electric potential produced by a pair of conventional AA batteries. This means the new invention can power devices that lower-voltage transient batteries cannot. However, use of lithium makes it unsuitable for biomedical applications such as power implants.



According to researchers at Iowa State University, USA, the battery can power a laptop for 15 minutes, but dissolves completely in water within 30 minutes (Image courtesy: www.indianexpress.com)

Machine learning to help eradicate world poverty

Scientists at Stanford University, USA, have found a low-cost method to map poverty in areas previously devoid of data, by combining satellite images and making use of machine learning. These improved poverty maps could help organisations and policymakers distribute funds more efficiently and evaluate policies.

One big challenge in providing relief to people living in poverty is locating them. Availability of accurate and reliable information on the location of impoverished zones is surprisingly lacking for much of the world, particularly in African countries. Aid groups and other international organisations often fill in the gaps with door-to-door

surveys, but these can be expensive and time-consuming to conduct.

Researchers at Stanford University have used machine learning, which is the science of designing computer algorithms that learn from data, to extract information about poverty from high-resolution satellite imagery.

The team built on earlier machine learning methods to find impoverished areas across five African countries—Nigeria, Tanzania, Uganda, Malawi and Rwanda. Because areas that are brighter at night are usually more developed, the solution involves combining high-resolution daytime imagery with images of Earth at night.